

# Meet In The Middle (M.I.T.M)

“ Concepts & Qns ”



Video-4



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Motivation .



You can't expect a top rank from average effort.

MIK

## 2035. Partition Array Into Two Arrays to Minimize Sum Difference

Hard

Topics

Companies

Hint

You are given an integer array `nums` of  $2 * n$  integers. You need to partition `nums` into **two** arrays of length `n` to **minimize the absolute difference** of the **sums** of the arrays. To partition `nums`, put each element of `nums` into **one** of the two arrays.

Return the **minimum** possible absolute difference.

Example 1:

|          |   |   |   |   |
|----------|---|---|---|---|
| array 1: | 3 | 9 |   |   |
| nums:    | 3 | 9 | 7 | 3 |
| array 2: |   |   | 7 | 3 |

Input: `nums = [3,9,7,3]`

Output: `2`

Explanation: One optimal partition is: `[3,9]` and `[7,3]`.

The absolute difference between the sums of the arrays is  $\text{abs}((3 + 9) - (7 + 3)) = 2$ .

Handwritten notes for Example 1:

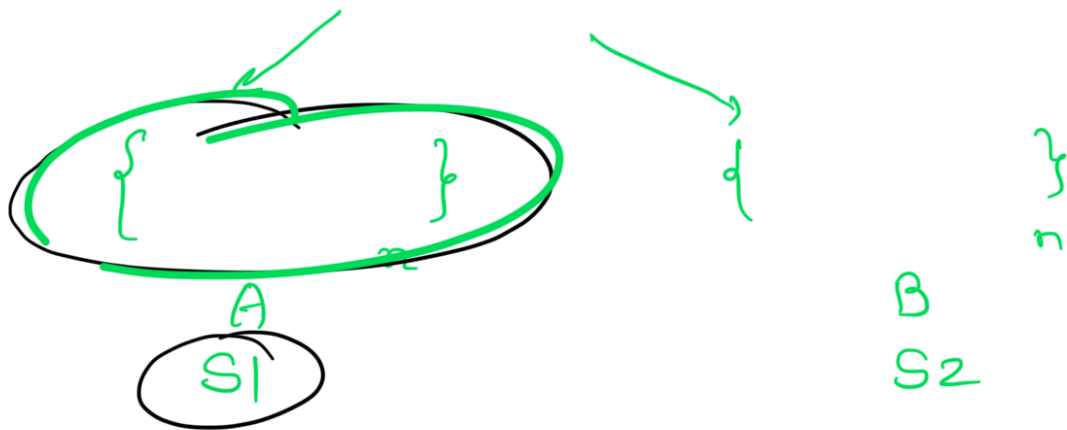
$\{3, 9, 7, 3\}$  (sum 22)  
 $\{3, 9\}$  (sum 12)  
 $\{7, 3\}$  (sum 10)  
Difference:  $12 - 10 = 2$

### Constraints:

- $1 \leq n \leq 15$
- `nums.length == 2 * n` (Handwritten: `== 30`)
- $-10^7 \leq \text{nums}[i] \leq 10^7$

# Thought Process...

nums = { , 2n



abs(s1-s2) minimize

$$\begin{aligned} S1 + S2 &= S & \text{--- (i)} \\ S2 &= (S - S1) & \text{--- (ii)} \end{aligned}$$

abs(s1-s2) → minimize

$$\text{abs}(s_1 - (s - s_1)) \rightarrow \text{using Eqn (ii)}$$

$$\boxed{\text{abs}(2 \cdot s_1 - s)}$$
 minimize

$$\{ \quad \quad \quad \}^A$$

We only have to worry about  $s_1(A)$ .

$S_1 \leadsto$  Sum of elements of array  $A$ .  
Size =  $n$

$$\text{nums} = \{a, b, c, \dots\}, 2n \approx 30$$

$\swarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$   
 $\checkmark \quad \times \quad \checkmark \quad \times \quad \checkmark$

$$A = \{ \quad \quad \quad \}, n$$

$\leftarrow$  DP (R/M/BUP)

$$\text{Possibl}_i = 2^{30} \approx 10^9 \times$$

Solve (i), count, sum)

TLE

$$\begin{matrix} [2n] & [ & ] & [ & ] \\ [30] & [15] & [1.5 \cdot 10^8] \end{matrix}$$

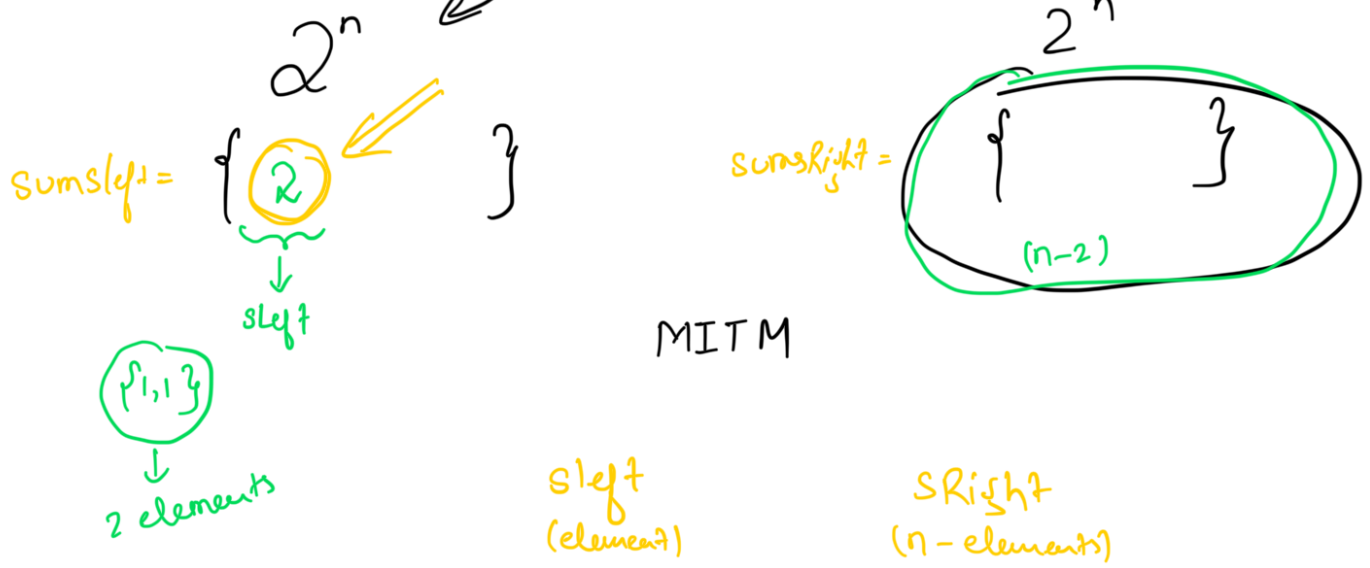
$$\begin{matrix} n * 10^7 \\ 15 * 10^7 \\ \downarrow \\ 1.5 * 10^8 \end{matrix}$$

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$$\begin{matrix} & 2^{2n} & \\ & \swarrow \quad \searrow & \\ 2^n & & 2^n \\ 2^{15} \approx 10^4 & & \end{matrix}$$

$$\text{nums} = \{ \quad \quad \quad \}, 2n$$

$$\text{Possible} = 2^{2n} \quad \times$$



$$\text{left} + \text{sRight} = S1$$

How to find Subset Sum  
using bit mask.

$\{ \}$   $2^n$

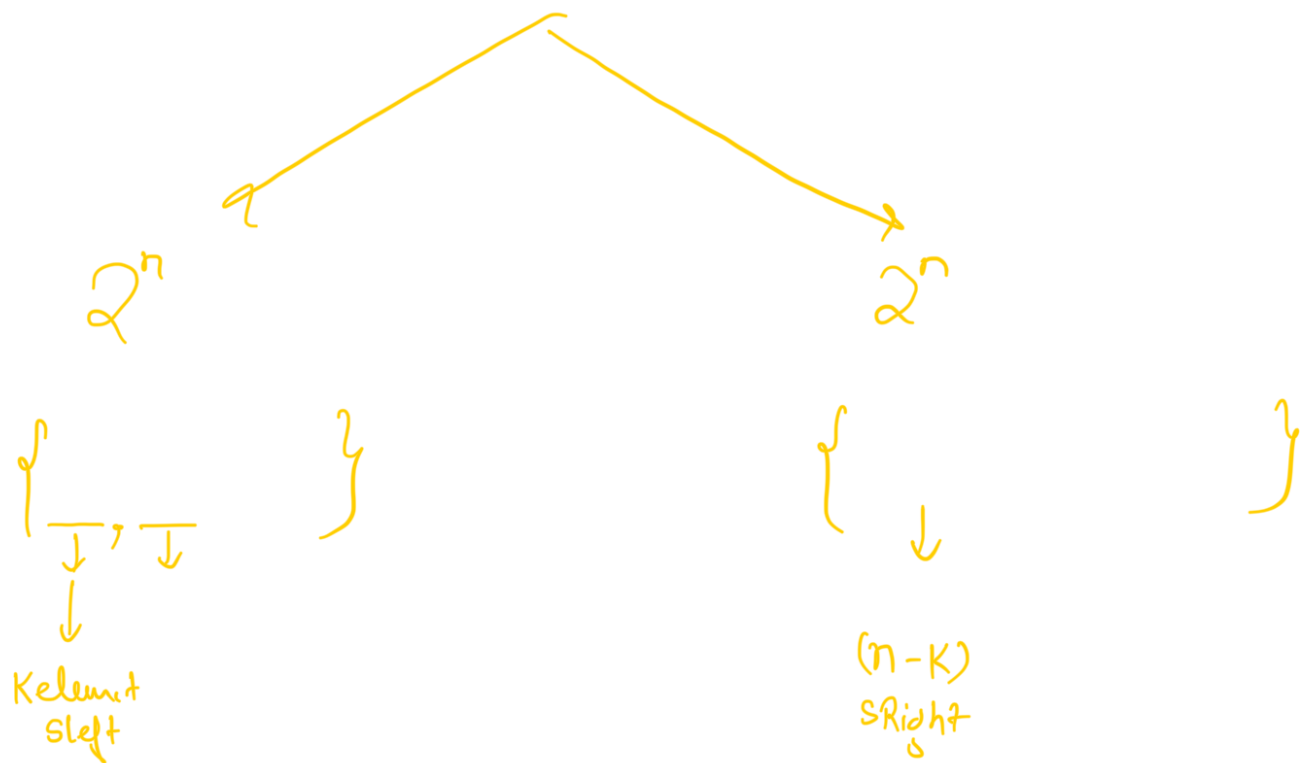
$2^n$  Possible Subset Sums

$\alpha$

$S1 \leadsto n \text{ elements}$



$$(2 \cdot S1 - S) \text{ ---}$$

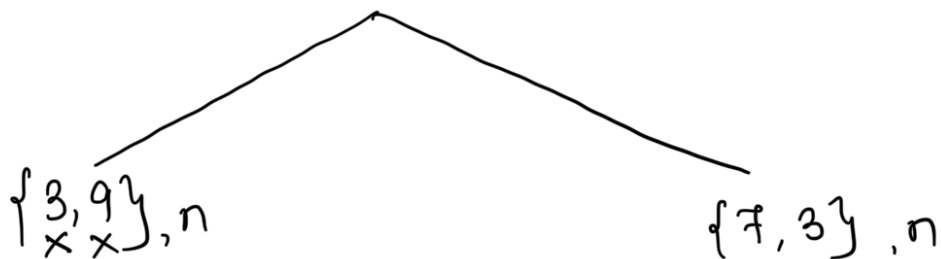


# Finding Subset Sums



along with count of  
elements in subset

nums = {<sup>0</sup>3, <sup>1</sup>9, <sup>2</sup>7, <sup>3</sup>3}, 2n



|          | mask<br><small>1111, 0111, 1111, 1111</small> | sum       |
|----------|-----------------------------------------------|-----------|
| { }      | <u>00</u>                                     | <u>0</u>  |
| { 3 }    | 01                                            | <u>3</u>  |
| { 9 }    | 10                                            | <u>9</u>  |
| { 3, 9 } | 11                                            | <u>12</u> |

map  
size → {subset sums}

0 → {0}

1 → {3, 9}

2 → {12}

|          | mask | sum      |
|----------|------|----------|
| { }      | 00   | 0        |
| { 7 }    | 01   | <u>7</u> |
| { 3 }    | 10   | <u>3</u> |
| { 7, 3 } | 11   | 10       |

map  
size → {subset sums}

0 → {0}

1 → {7, 3} → {3, 7}

2 → {10}



K=1 size

→ {3, 9}

(n-K) size

{7, 3}

Brute force.

$$\text{abs}(2 \cdot s1 - S)$$

$$S = 22$$

$$s1 = 3 + 7 = \underline{10}$$

2

$$s1 = 3 + 3 = \underline{6}$$

10

$$s1 = 9 + 7 = \underline{16}$$

10

$$s1 = 9 + 3 = \underline{12}$$

2

$$s1 = \underset{\downarrow}{3} + \underset{\downarrow}{7} = 10$$

{8} {7}

{3, 7}

{3, 9}

$$\text{abs}(2 \cdot s1 - S)$$

MITM...

map subset Sums

Size  $\rightarrow \{$

0  $\rightarrow \{0\}$

1  $\rightarrow \{3, 9\}$

2  $\rightarrow \{12\}$

map

size  $\rightarrow \{ \text{subset Sums} \}$

0  $\rightarrow \{0\}$

1  $\rightarrow \{3, 7\}$

2  $\rightarrow \{10\}$   $> 2$

K=1

$n-K = 2-1=1$

$\{3, 9\}$

leftSubSum = 9

rightSubSum = ?

$\{9\}$

$\{3\}$

$\{9, 3\}$

$\rightarrow S1$

$$\text{abs}(2 \cdot S1 - S) \text{ min} \rightarrow 0$$

$$\text{abs}(2 \cdot (\text{leftSubSum} + \text{rightSubSum}) - S)$$

$$(\text{leftSubSum} + \text{rightSubSum}) \approx \frac{S}{2}$$

$$\text{rightSubSum} = \frac{S}{2} - \text{leftSubSum}$$

$$\text{rightSubSum} = \left( \frac{S - 2 \cdot \text{leftSubSum}}{2} \right)$$

$$= \frac{22 - 18}{2}$$

$$= 4/2$$

$$= 2 \longrightarrow$$

lower-bound = 3

$$2fs = 9$$

$$\text{abs}(2 \cdot S1 - S)$$

$$\text{abs}(2 \cdot (9 + 3) - 22)$$

$$\text{abs}(24 - 22) = \underline{\underline{2}}$$

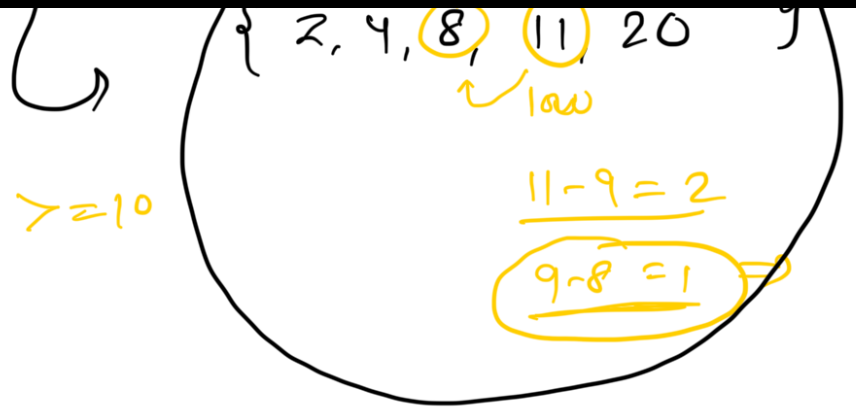
Find near value of 9

map

0 → { }

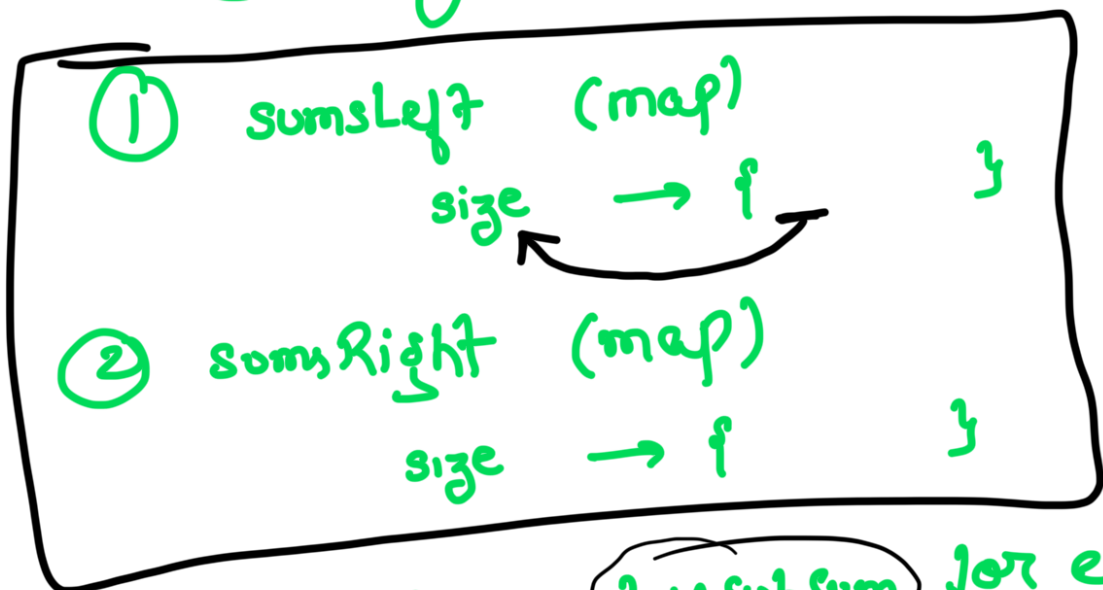
1 → { }

2 → { }



Video-3.

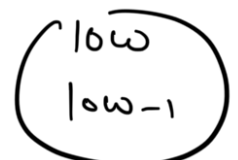
## Story Points.



③ left side  $\rightarrow$  leftSubSum for each size (K)

④ right side  $(n-K) \rightarrow$  lower-bound  
 $rightSum = \left( \frac{s - 2 \cdot leftSum}{2} \right)$

⑤  $abs(2 \cdot s1 - s)$



n

```
for (int mask = 0 ; mask < (1 << n) ; mask++) {
```

```
    int S = 0;  
    int count = 0;
```

```
    for (i = 0; i < n; i++) {  
        if (mask & (1 << i)) {  
            S += nums[i];  
            count++;  
        }  
    }
```

```
    sumsLeft[count].push-back(S);
```

~~sumsRight~~ (n)

Sort

```
S += nums[n+i];
```